BEE 4750 Homework 3: Dissolved Oxygen and Monte Carlo

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Due Date

Thursday, 10/16/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to implement a model for dissolved oxygen in a river with multiple waste releases and use this to develop a strategy to ensure regulatory compliance.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `~/hw3-claire`
```

```
using Random
using Plots
using LaTeXStrings
using Distributions
```

Problems (Total: 30 Points)

Problem 1 (30 points)

A river which flows at 6 km/d is receiving waste discharges from two sources which are 15 km apart. The oxygen reaeration rate is 0.55 day-1, and the decay rates of CBOD and NBOD are are 0.35 and 0.25 day-1, respectively. The river's saturated dissolved oxygen concentration is 10m g/L.

If the characteristics of the river inflow and waste discharges are given in Table 1, write a Julia model to compute the dissolved oxygen concentration from the first wastewater discharge to an arbitrary distance `d` km downstream. Use your model to compute the minimum dissolved oxygen concentration up to 50 km downstream and how far downriver this maximum occurs.

Parameter	River Inflow	Waste Stream 1	Waste Stream 2
Inflow	100,000 m3/d	10,000 m3/d	15,000 m3/d
DO Concentration	7.5 mg/L	5 mg/L	5 mg/L
CBOD	5 mg/L	50 mg/L	45 mg/L
NBOD	5 mg/L	35 mg/L	35 mg/L

Table 1: River inflow and waste stream characteristics for Problem 1.

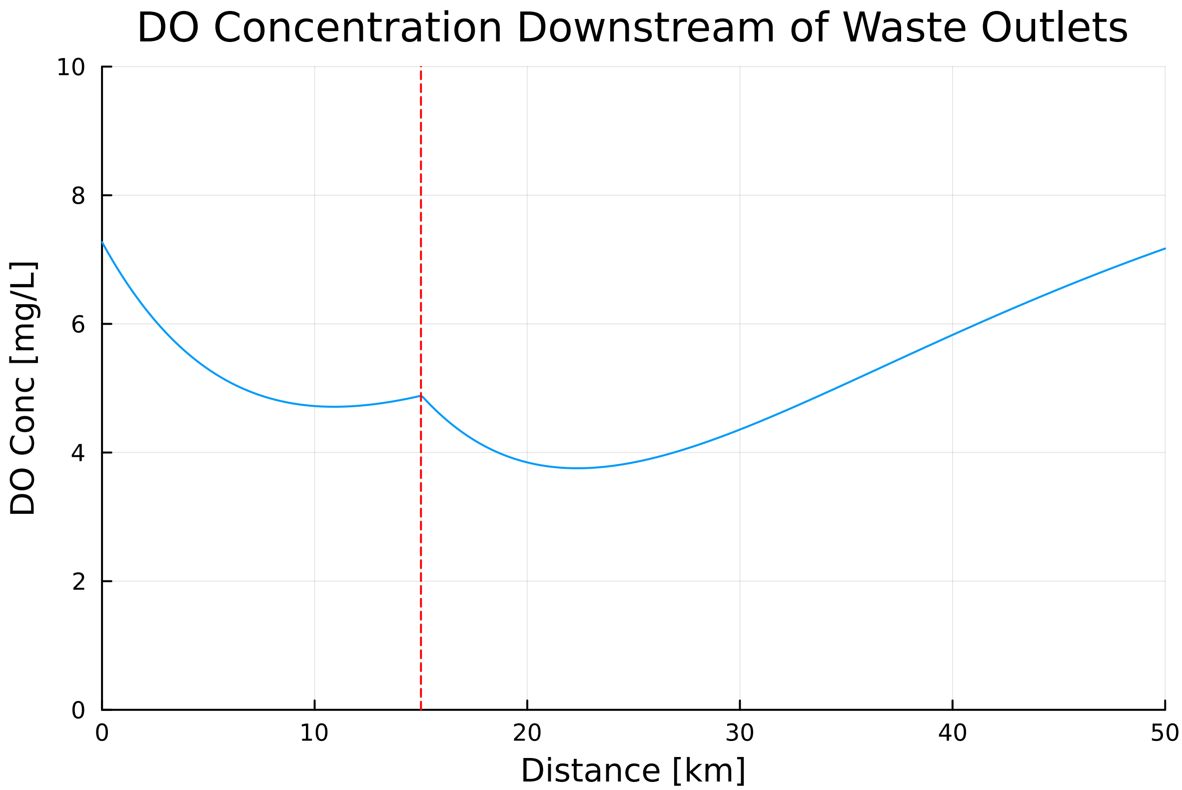
Problem 1.1

Implement the Streeter-Phelps (analytic) solution for the dissolved oxygen concentration. Plot the dissolved oxygen concentration from the first waste stream to 50 km downriver. What is the minimum value in mg/L?

```
x2 = 15 # km
xf = 50 # km
dx = 0.1 # km
u = 6 # km/d

C_sat = 10 # mg/L

kr = 0.55 # day^-1
kc = 0.35 # day^-1
kn = 0.25 # day^-1

Qr = 100000 # m^3/day
Cdo_r = 7.5 # mg/L
```

```
Cc_r = 5 # mg/L
Cn_r = 5 # mg/L

Q1 = 10000 # m^3/day
Cdo_1 = 5 # mg/L
Cc_1 = 50 # mg/L
Cn_1 = 35 # mg/L

Q2 = 15000 # m^3/day
Cdo_2 = 5 # mg/L
Cc_2 = 45 # mg/L
Cn_2 = 35 # mg/L

#find new cconcentration when two streams mix

C_new(q1, c1, q2, c2) = (q1 * c1 + q2 * c2) / (q1 + q2)

#get total bod from rates, cbod, and nbod
function mix(DO_0, CBOD0, NBOD0, t, k1, k2, k3, sat)
    DO_new = DO_0 * exp(-k1 * t)
    CBOD_new = CBOD0 * (k2 / (k1 - k2)) * (exp(-k2 * t) - exp(-k1 * t))
    NBOD_new = NBOD0 * (k3 / (k1 - k3)) * (exp(-k3 * t) - exp(-k1 * t))
    D = DO_new + CBOD_new + NBOD_new
    return sat - D
end

#get do from when the first waste effluent first enters the stream
Q_first = Qr + Q1
DO_mix1 = C_new(Qr, Cdo_r, Q1, Cdo_1)
CBOD_mix1 = C_new(Qr, Cc_r, Q1, Cc_1)
NBOD_mix1 = C_new(Qr, Cn_r, Q1, Cn_1)
DO_first = C_sat - DO_mix1

#decay equation to apply various q values
decay(C_0, k, t) = C_0 * exp(-k * t)

xs = 0.0:dx:xf
DOs= zeros(length(xs))

for (i, x) in enumerate(xs)
    #DO before the second effluent comes in
    if x <= x2
```

```
        t = x / u
        DOs[i] = mix(DO_first, CBOD_mix1, NBOD_mix1, t, kr, kc, kn, C_sat)
    else
        #find DO right at second effluent and get change over time after that
        t_x2 = x2 / u
        DO_x2 = mix(DO_first, CBOD_mix1, NBOD_mix1, t_x2, kr, kc, kn, C_sat)
        CBOD_x2 = decay(CBOD_mix1, kc, t_x2)
        NBOD_x2 = decay(NBOD_mix1, kn, t_x2)
        DO_mix2 = C_new(Q_first, DO_x2, Q2, Cdo_2)
        CBOD_mix2 = C_new(Q_first, CBOD_x2, Q2, Cc_2)
        NBOD_mix2 = C_new(Q_first, NBOD_x2, Q2, Cn_2)
        DO_second = C_sat - DO_mix2
        t_after2 = (x - x2) / u
        DOs[i] = mix(DO_second, CBOD_mix2, NBOD_mix2, t_after2, kr, kc, kn, C_sat)
    end
end

#find the minimum dissolved oxygen concentration
min_conc=minimum(DOs)

#plot concentration over time, second effluent location marked
p = plot(xs, DOs,
    xlim = (0, 50),
    ylim = (0,10),
    legend = false,
    xlabel = "Distance [km]",
    ylabel = "DO Conc [mg/L]",
    title = "DO Concentration Downstream of Waste Outlets"
)

vline!(p, [15], color=:red, linestyle=:dash)

display(p)
println("Minimum dissolved oxygen concentration = ", min_conc)
```




```
Minimum dissolved oxygen concentration = 3.75570046982907
```

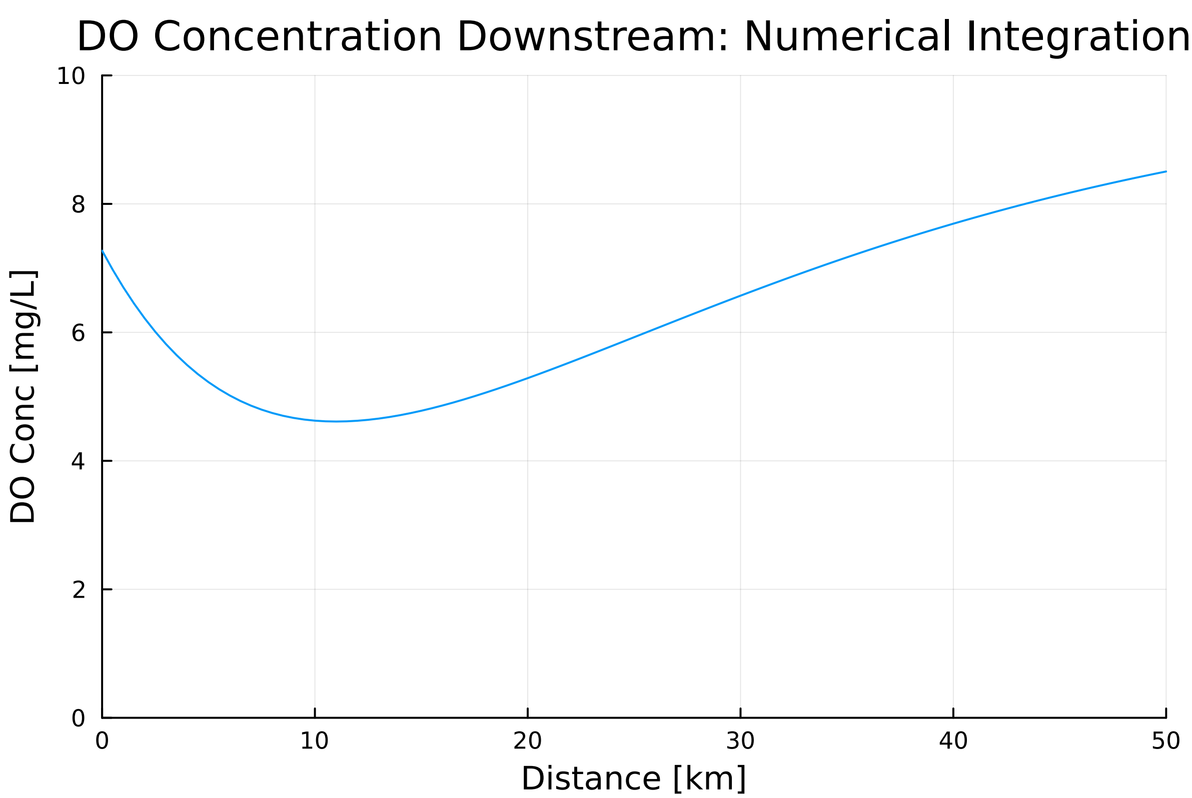
Problem 1.2

Implement a numerically-integrated (discretized) solution for the dissolved oxygen concentration. Using a resolution of 0.5km, conduct a simulation and plot the results on the same axis as the plot from Problem 1.1. How has the minimum value changed? What do you attribute this difference to (be specific about the source of the difference(s) in terms of the simulation dynamics, not just that one is a numerical approximation).

The new minimum is ~0.9 mg/L higher than in the analytic solution. This can likely be attributed to some truncation error from the larger time step as compared to the analytical solution and the fact that the values are only incrementally evaluated. Also, the numerical integration method assumes constant values and only has values compared to those 0.5 km before, ignoring what goes on in between those time steps meaning a minimum will be mostly ignored if it lies between two time steps. This means a change could be ignored even further as the effects of evaluating based on that point will also be ignored.

```
#set resolution
dx_int = 0.5 # km
xs_int = 0.0:dx_int:xf
dt_int = dx_int / u

#initialize arrays for DO, CBOD, and NBOD for iterations
DO_int=zeros(length(xs_int))
CBOD_int = zeros(length(xs_int))
NBOD_int = zeros(length(xs_int))

#set first value for each array
DO_int[1] = DO_mix1
D=C_sat-DO_mix1
CBOD_int[1] = CBOD_mix1
NBOD_int[1] = NBOD_mix1

#fill in rest of array by evaluating numerical integration
for i in 2:length(xs_int)
    CBOD_int[i] = CBOD_int[i-1] * exp(-kc*dt_int)
    NBOD_int[i] = NBOD_int[i-1] * exp(-kn*dt_int)

    #calculating new deficit
    D = D * exp(-kr*dt_int) + (CBOD_int[i-1] - CBOD_int[i])+(NBOD_int[i-1]-NBOD_int[i])

    #next do concentration from saturation value and deficit
    DO_int[i] = C_sat - D

    #get value as close to second outlet as possible
    if xs_int[i-1] < x2 < xs_int[i]
        DO_int[i] = C_new(Q_first, DO_int[i], Q2, Cdo_2)
        CBOD_int[i] = C_new(Q_first, CBOD_int[i], Q2, Cc_2)
        NBOD_int[i] = C_new(Q_first, NBOD_int[i], Q2, Cn_2)
        DO= C_sat - DO_int[i]
        Q_first += Q2
    end
end

#get the minimum for the integrated set
min_conc_new=minimum(DO_int)

#make new plot of new do concentrations with same axes as previous
p2 = plot(xs_int, DO_int,
```

```
    xlim = (0, 50),
    ylim = (0,10),
    legend = false,
    xlabel = "Distance [km]",
    ylabel = "DO Conc [mg/L]",
    title = "DO Concentration Downstream: Numerical Integration"
)

display(p2)
println("Minimum dissolved oxygen concentration = ", min_conc_new)
```




```
Minimum dissolved oxygen concentration = 4.612126754365276
```

Problem 1.3

Using the analytic model, what is the minimum level of treatment (% removal of organic waste; assume this is equivalent to the same level of reduction of the CBOD and NBOD) for waste stream 1 that will ensure that the dissolved oxygen concentration is in compliance with the

4 mg/L standard along with a 5% margin of safety, assuming that waste stream 2 remains untreated? How about if only waste stream 2 is treated?

```
#return to original time step
dx=0.1
x2=15
xf=50
u=6

function find_treat(tr1, tr2)
    #change input concentrations based on treatment percentages
    Cc_1_t = Cc_1*(1-tr1) # mg/L
    Cn_1_t = Cn_1*(1-tr1) # mg/L

    Cc_2_t = Cc_2*(1-tr2)
    Cn_2_t = Cn_2*(1-tr2)

    #reevaluate initial DO concentrations based on treated effluent
    Q_first = Qr + Q1
    DO_mix1 = C_new(Qr, Cdo_r, Q1, Cdo_1)
    CBOD_mix1_t = C_new(Qr, Cc_r, Q1, Cc_1_t)
    NBOD_mix1_t = C_new(Qr, Cn_r, Q1, Cn_1_t)
    DO_first = C_sat - DO_mix1

    decay(C_0, k, t) = C_0 * exp(-k * t)

    xt = 0.0:dx:xf
    DOs_t = zeros(length(xt))

    #same analytical method as before
    for (i, x) in enumerate(xt)
        if x <= x2
            t = x / u
            DOs_t[i] = mix(DO_first, CBOD_mix1_t, NBOD_mix1_t, t, kr, kc, kn, C_sat)
        else
            t_x2 = x2 / u
            DO_x2_t = mix(DO_first, CBOD_mix1_t, NBOD_mix1_t, t_x2, kr, kc, kn, C_sat)
            CBOD_x2_t = decay(CBOD_mix1_t, kc, t_x2)
            NBOD_x2_t = decay(NBOD_mix1_t, kn, t_x2)
            DO_mix2_t = C_new(Q_first, DO_x2_t, Q2, Cdo_2)
            CBOD_mix2_t = C_new(Q_first, CBOD_x2_t, Q2, Cc_2_t)
            NBOD_mix2_t = C_new(Q_first, NBOD_x2_t, Q2, Cn_2_t)
            DO_second_t = C_sat - DO_mix2_t
```

```
            t_after2_new = (x - x2) / u
            DOs_t[i] = mix(DO_second_t, CBOD_mix2_t, NBOD_mix2_t, t_after2_new, kr, kc, kn, C
        end
    end

    # Compute new minimum concentration
    min_conc = minimum(DOs_t)
    return min_conc
end
```

```
find_treat (generic function with 1 method)
```

```
C_std = 4*(1.05)

#Finding percentage treatment on stream 1 which keeps min. conc. up to standard
for r1 in 0:0.01:1
    min_conc = find_treat(r1, 0)
    if min_conc >= C_std
        println("Stream 1 treatment required = ", round(r1*100, digits=1), "%")
        break
    end
end

# Finding percentage treatment on stream 2 keeping min. conc. up to standard
for r2 in 0:0.01:1
    min_conc = find_treat(0, r2)
    if min_conc >= C_std
        println("Stream 2 treatment required = ", round(r2*100, digits=1), "%")
        break
    end
end
```

```
Stream 1 treatment required = 26.0%
Stream 2 treatment required = 20.0%
```

Problem 1.4

Suppose you are responsible for designing a waste treatment plan for discharges into the river, with a regulatory mandate to keep the dissolved oxygen concentration above 4 mg/L. Discuss whether you'd opt to treat waste stream 2 alone or both waste streams equally. What other information might you need to make a conclusion, if any?

I would prefer to treat stream both equally. Although stream 2 requires less percentage of treatment, treating both streams would limit the effects throughout the entire stream. When only the second effluent is treated it is more of a reactive approach than proactive, allowing the 15 km stretch of the river to receive all of the waste from the first effluent. In order to make a more accurate conclusion I would like to know the difference in cost of the two strategies (will that much money really be saved by treating only outlet 2). I would also like to know how much the BOD concentrations fluctuate. Having both outlets treated allows to prepare for increases in input from either one.

Problem 1.5

Suppose that it is known that the DO concentrations at the river inflow can vary according to a LogNormal$(2.0, 0.15)$ distribution. Conduct a Monte Carlo simulation of the DO concentration in the river. If you treat only waste stream 1 (based on your analysis from Problem 1.3), what is the expected probability and 95% confidence interval that the river fails to comply with the regulatory standard of 4 mg/L? How did you decide that your Monte Carlo sample size was sufficiently large?

```
sample_size=1000000
# inflow DO LogNormal distribution
min_vals = zeros(sample_size)
DO_r__logn = rand(LogNormal(2.0, 0.15), sample_size)

xs = 0.0:dx:xf

Cc_1_t=Cc_1*(1-0.26)
Cn_1_t=Cn_1*(1-0.26)

#evaluating total concentration for each initial conc. from the distribution
for i in 1:sample_size
    Cdo_r = DO_r__logn[i]

    Q_first = Qr + Q1
    DO_mix1 = C_new(Qr, Cdo_r, Q1, Cdo_1)
    CBOD_mix1 = C_new(Qr, 0.0, Q1, Cc_1_t)
    NBOD_mix1 = C_new(Qr, 0.0, Q1, Cn_1_t)
    DO_first = C_sat - DO_mix1

    #array of same size
    DOs = similar(xs)

    for (j, x) in enumerate(xs)
```

```
            t = x / u
            DOs[j] = mix(DO_first, CBOD_mix1, NBOD_mix1, t, kr, kc, kn, C_sat)

    end

    #add each minimum conc. for evaluation compared to regulation
    min_vals[i] = minimum(DOs)
end

failures = sum(min_vals .< 4.0)
p_fail = failures / sample_size

z =1.96
se = sqrt(p_fail * (1 -p_fail) /sample_size)
ci_lower=p_fail-z*se
ci_upper=p_fail+z*se

println("Estimated probability of non-complianc: ", round(p_fail, digits=4))
println("95% Confidence Interval: [", round(ci_lower, digits=4), ", ", round(ci_upper, digits

sample_i = rand(1:sample_size)
Cdo_r = DO_r__logn[sample_i]
Q_first = Qr + Q1
DO_mix1 = C_new(Qr, Cdo_r, Q1, Cdo_1)
CBOD_mix1 = C_new(Qr, 0.0, Q1, Cc_1)
NBOD_mix1 = C_new(Qr, 0.0, Q1, Cn_1)
DO_first = C_sat - DO_mix1
DOs = [mix(DO_first, CBOD_mix1, NBOD_mix1, x/u, kr, kc, kn, C_sat) for x in xs]

plot(xs, DOs,
    xlim=(0, 50),
    ylim=(0, 10),
    legend=false,
    xlabel="Distance [km]",
    ylabel="DO [mg/L]",
    title="Example DO Profile (Random Trial)")
vline!([15], color=:red, linestyle=:dash)
```

References

List any external references consulted, including classmates.